Lecture 18 Scribe

L18.pdf
PDF

okay so i have to generate a lecture scribe for this lecture (L18), and this scribe is for exam preparation, like if someone studies only this they should be able to revise properly and recall what was taught in class , while writing, you should reason step by step through everything, like definitions, assumptions, proofs, derivations and examples, and this reasoning should also be visible in the scribe so that the logical flow is clear while studying , but all of this must be strictly based only on the provided lecture slides/pdf and textbook content, don't use anything from outside, and don't add any new explanation, intuition or examples which are not present in the given material

include all important things like definitions, notation, assumptions, theorem statements, and wherever proofs or derivations are there, write them properly step by step exactly as in lecture so that they can be followed during revision, same for examples also

also follow the same structure and order as the lecture, so the flow of topics and ideas remains consistent and easy to understand later

importantly..... do not invent or assume anything which is not explicitly given in the context

overall it should look like proper academic exam notes, not casual explanation, and reasoning should help in understanding the steps as taught in lecture only

Below is your **COT-style Lecture 18 Scribe** strictly based on the provided slides .
No external additions, no extra intuition—only structured, step-by-step reasoning exactly as in lecture.

CSE400 – Lecture 18 Scribe

Marginal PMF, Correlation, Covariance, Joint Gaussian RVs

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1. Marginal PMF

1.1 Definition

For discrete random variables X and Y, the joint PMF is:

$$p_{XY}(x, y) = P(X = x, Y = y)$$

Derivation of Marginal PMF

To obtain the marginal PMF of one variable:

- For X:

$$p_X(x) = \sum_y p_{XY}(x, y)$$

- For Y:

$$p_Y(y) = \sum_x p_{XY}(x, y)$$

Key Idea (as stated in lecture)

The marginal PMF of one variable is obtained by **summing the joint PMF over all possible values of the other variable**.

1.2 Marginal PMF from Joint PMF Table

From the table representation (page 7–8):

- **Row sums** → $p_X(x)$

$$p_X(x_i) = \sum_j p_{ij}$$

- **Column sums** → $p_Y(y)$

$$p_Y(y_j) = \sum_i p_{ij}$$

Also,

$$\sum_i p_X(x_i) = 1, \quad \sum_j p_Y(y_j) = 1$$

1.3 Example

Given joint PMF table:

	$y = 0$	$y = 1$
$x = 0$	$1/8$	$1/4$
$x = 1$	$3/8$	$1/4$

Step-by-step solution

Compute $p_X(x)$

$$p_X(0) = \frac{1}{8} + \frac{1}{4} = \frac{3}{8}$$

$$p_X(1) = \frac{3}{8} + \frac{1}{4} = \frac{5}{8}$$

Compute $p_Y(y)$

$$p_Y(0) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}$$

$$p_Y(1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Final Marginals

$$p_X(x) = \begin{cases} \frac{3}{8}, & x = 0 \\ \frac{5}{8}, & x = 1 \end{cases} \quad p_Y(y) = \begin{cases} \frac{1}{2}, & y = 0 \\ \frac{1}{2}, & y = 1 \end{cases}$$

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2. Correlation

2.1 Definition

A basic measure of joint behavior:

$$R_{XY} = E[XY]$$

Interpretation

- Represents the **average value of the product** XY
- Called a **raw correlation measure**

2.2 Observations

- Depends on scale of X and Y
- Computed **about origin (not mean)**
- Combines:
  - mean values
  - spread
  - dependence

Important Result

$$R_{XY} = 0 \Rightarrow \text{orthogonal about origin}$$

Transition

To remove effect of means → **Covariance**

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3. Covariance

3.1 Definition

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Where:

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

3.2 Interpretation

Steps (as per lecture reasoning):

1. Subtract mean from each variable
2. Measure joint variation of centered variables

Meaning

- Positive → deviations move in same direction
- Negative → deviations move in opposite direction

- Zero → no linear co-variation

Important Note

Covariance:

- Gives **direction of linear relationship**
- Magnitude depends on units → **cannot measure strength**

3.3 Algebraic Form

Start with:

$$\mathrm{Cov}(X,Y) = E[(X-\mu_X)(Y-\mu_Y)]$$

Expand:

$$= E[XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y]$$

Apply linearity:

$$= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X\mu_Y$$

Since:

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

Therefore:

$$\mathrm{Cov}(X,Y) = E[XY] - \mu_X\mu_Y$$

Final form:

$$\boxed{\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]}$$

3.4 Properties

1. **Symmetry**

$$\mathrm{Cov}(X,Y) = \mathrm{Cov}(Y,X)$$

2. **Variance as Special Case**

$$\mathrm{Cov}(X,X) = \mathrm{Var}(X)$$

3. **Linearity**

$$\mathrm{Cov}(aX+b, cY+d) = ac \cdot \mathrm{Cov}(X,Y)$$

4. **Independence**

$$X \perp Y \Rightarrow \mathrm{Cov}(X,Y) = 0$$

⚠️ Important:

- Zero covariance ⇒ **uncorrelated**
- Does **not imply independence**

3.5 Example 1

Given:

$$Y = -2X + 3$$

Find:

$$\mathrm{Cov}(X, Y)$$

Step 1: Use formula

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Step 2: Compute components

$$XY = X(-2X + 3) = -2X^2 + 3X$$

$$E[Y] = E[-2X + 3] = -2E[X] + 3$$

Step 3: Substitute

$$\begin{aligned}\mathrm{Cov}(X, Y) &= E[-2X^2 + 3X] - E[X](-2E[X] + 3) \\ &= -2E[X^2] + 3E[X] + 2(E[X])^2 - 3E[X] \\ &= -2E[X^2] + 2(E[X])^2\end{aligned}$$

Step 4: Use variance identity

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

Thus:

$$\mathrm{Cov}(X, Y) = -2\mathrm{Var}(X)$$

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4. Pearson's Correlation Coefficient

4.1 Definition

$$\rho_{XY} = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

4.2 Properties

- Measures **strength and direction**
- Normalized covariance
- Unit-free

Range:

$$-1 \leq \rho_{XY} \leq 1$$

Interpretation:

- > 0: positive linear trend
- < 0: negative linear trend
- $= 0$: no linear correlation

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5. Covariance – Example 2 (Continuous Case)

Given joint PDF:

$$f_{XY}(x,y) = \begin{cases} \frac{xy}{96}, & 0 < x < 4,\ 1 < y < 5 \\ 0, & \text{otherwise} \end{cases}$$

Step 1: Compute $E[X]$

$$E[X] = \int_0^4 x f_X(x)\, dx = \frac{1}{8}\int_0^4 x^2 dx$$
$$= \frac{1}{8} \cdot \frac{64}{3} = \frac{8}{3}$$

Step 2: Compute $E[Y]$

$$E[Y] = \frac{1}{12}\int_1^5 y^2 dy = \frac{31}{9}$$

Step 3: Compute $E[XY]$

Since X and Y are independent:

$$E[XY] = E[X]E[Y] = \frac{8}{3} \cdot \frac{31}{9} = \frac{248}{27}$$

Step 4: Compute $E[2X + 3Y]$

$$E[2X + 3Y] = 2E[X] + 3E[Y]$$
$$= 2 \cdot \frac{8}{3} + 3 \cdot \frac{31}{9} = \frac{47}{3}$$

Step 5: Compute $\mathrm{Var}(X)$

$$E[X^2] = 8$$

$$\mathrm{Var}(X) = 8 - \left(\frac{8}{3}\right)^2 = \frac{8}{9}$$

Step 6: Compute $\mathrm{Var}(Y)$

$$E[Y^2] = 13$$

$$\mathrm{Var}(Y) = 13 - \left(\frac{31}{9}\right)^2 = \frac{92}{81}$$

Step 7: Compute Covariance

$$\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0$$

Step 8: Compute Correlation

$$\rho_{XY} = 0$$

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6. Jointly Gaussian Random Variables

6.1 Definition

A pair (X, Y) is **jointly Gaussian** if:

- Joint density is **bivariate Gaussian**
- Marginals are Gaussian

6.2 Joint Gaussian PDF

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\right.\right.$$

Where:

- μ_X, μ_Y: means
- σ_X, σ_Y: standard deviations
- ρ: correlation coefficient

6.3 Interpretation (from slides)

- Used in:
  - signal processing
  - communication systems
  - noisy measurements
- Example (given in lecture):
  - X: CO_2 concentration
  - Y: temperature anomaly
  - Joint behavior approximated as Gaussian
- If:

$$\rho > 0$$

→ higher CO_2 associated with higher temperature

✅ End of Scribe (Strict Lecture-Based Notes)

If you want next step: I can convert this into **perfect exam revision cheat sheet (1–2 pages)** or **last-day revision flowchart**.